Uavx-nanoV4

Figure 1: Uavx-nanoV4

Introduction

This is the evolving setup page for the Arm version of UAVX. We will try to keep explanations concise. If you have flown UAVX then most of this is familiar as we have deliberately scaled the parameters and kept the look and feel of UAVPSet and UAVXGS to those you are familiar with so you can concentrate on flying.

A Python GCS reimplementation (UAVXGS, in `uavx-python/src/`) is available alongside the legacy Java UAVXGUI and UAVPSet. It provides parameter management, telemetry display, and mission planning. All references to UAVXGUI below apply equally to the Python GCS unless noted.

UAVXArm32F4 supports many different airframe types from octocopters, through helis out to conventional aircraft. UAVX has done this for several years. For fixed wing aircraft see the UAVXFixedWing supplemental wiki after reading this.

UAVXArm32F4 Board You should receive your board loaded with firmware compatible with UAVXGUI. You definitely should look at the UAVXGUI before you go any further. You will be able to load firmware updates later - see the UAVXArm32F4LoadingFirmware Wiki.

Step 1

Mount the board on your favourite frame oriented in the direction of forward flight. So if you are flying +Mode point the board along the K1 motor arm. If you are flying XMode point the board between the K1 and K3 motor arms. Select the desired configuration in UAVPSet using the airframe pulldown. We will assume from now that you are flying X-Mode which has become the most common as it allows cameras a clear forward view.

Take all the the shorting links (if any) off the board. Connect up an arming switch between the Arming pin and the adjacent Ground pin.

The motors should not run even if you have the arming switch on until you get through all of the following steps but think about a Kw of motors all running at around 7000RPM flying around your workshop - take the props OFF.

The board can be connected directly to your main battery (3S LiPo).

The Red light should be on and the Yellow LED should be flashing once a second indicating that you have not yet calibrated your accelerometers.

Step 2

Download the software package from the downloads area and install UAVXGUI.

Connect the adapter lead to the board and then to a USB port on your computer.

Start UAVXGUI and check that you have the appropriate COM port selected and that the baud rate is set to 115Kbaud.

Select the parameters button and read the current parameters which will be the defaults.

The default parameters were obtained from flights by Ken & Jim. These are a good starting point but you will probably wish to tune the Roll/Pitch and Yaw parameters in particular to suit your own flying style and the size/span and weight of your aircraft.

Step 3 (Rx)

From now on any changes you make to parameters in UAVXGUI must be followed by selecting the **Write Config** icon. Some parameter changes involve an electrical reconfiguration of the board so you will occasionally see all LEDs flashing - cycle the power and this should allow the board to complete the reconfiguration.

Connect a 5V supply to any of the M or Rx centre pins. This will power your Rx as centre pins of all of the M and Rx pins are connected together. In the past this would have been ONE of the ESC BECs. More recently a seperate 5V switching UBEC is used.

If you have a a Rx (e.g. FrSky) that supports Compound PPM (CPPM) connect it to Rx1. For CPPM you must specify the number of channels your Rx is receiving using UAVXGUI.

If you have an Rx that uses parallel PPM then connect it to Rx1 upwards USUALLY in the order Throttle, Aileron, Elevator, Rudder, Gear, Aux1, Aux2 etc.. You need a minimum of 7 channels for full functionality.

For parallel PPM using UAVXGUI set the number of Rx channels you have actually connected to a maximum of 8. If you are using a CPPM or UART connected Rx protocol such as as SBus (Preferred) then more channels are available.

You can fly with a minimum of 4 channels Throttle, Aileron, Elevator, Rudder. If you do so then you will have altitude hold but no navigation capability.

You may re-assign the channels using the selectors in UAVXGUI. So if your throttle is on Channel 3 for example then setup the pulldowns appropriately.

Some channels have multiple functions. You should read UAVXFlightModes.

Go to the main UAVXGUI page and turn on your Tx. Stir the Tx sticks and adjust the endpoints and neutral values of those channels that turn orange.

The Green LED should be on and Yellow LED flashing every second.

Step 4 (Acc/Gyro Temperature Calibration)

This is the most critical step in the setup and UAVX will not arm until it is done. You usually only need to do it once or until you next bend the aircraft so take your time.

Put the aircraft on a level surface. If it is on the slightest of angles then this is the way it will fly when you centralise the sticks. Bear in mind the floors of your house will not be level, your best table will not be level nor will your average workbench. Eyeballing it is no good at all.

Get a decent spirit level and even then turn it around to make sure it reads level both ways.

Lock the aircraft down so it does not move and recheck it is level.

Power up with UAVGUI connected. Select the CalIMU button which will be Red. You will see the blue LED start to flash - it has already taken the first set of readings and is waiting for the temperature to rise to take the second set. Get a hair dryer and VERY SLOWLY warm up the board - put your hand close to the board to sense how hot the air is. If it is too hot for your hand it is too hot for the board! You only need to change its temperature by 20 Celsius all up so don't melt the board! There is a delay as the heat gets into the MPU6xxx package so slowly/slowly until the blue LED stops flashing and you are done. The CalIMU button should go Green. Obviously it is better to do the calibration so it covers the range of temperatures you are likely to be flying in.

Once again you only need to do this once or when loading the defaults or if you change the gyro selection pulldown - it should be on UAVXArm32 unless you are using analog gyros.

The Green LED and Yellow LEDs should be on.

Optional Additional 6 Point Accelerometer Calibration Accelerometers may have slightly different sensitivity on their three axes. This means that while the zero angle measurement may be OK angles away from zero may be out by a few degrees which in most practical cases does not matter. You may choose to do an optional 6 point calibration which will determine the sensitivity scaling factors. Select the Acc6Pt button and the Blue LED only should be on. Then position the aircraft in the six possible primary orientations. For a quadcopter this will be top up, top down, left arm(s) down, right arms down, front arms down, rear arms down. These are all at 90 degrees to each other +/- 10 degrees so try to be precise. Do this as smoothly as possible so you minimise the shakes when you get to the measurement position although several hundred readings are taken in each position and the average used! For a fixed wing aircraft this is flat, inverted, nose up, nose down, left wing down, right wing down. The order you do this is not important but you must do all six directions.

If the orientation is correct you will get a green LED for about a second then red indicating you should move to the next position. If you see a yellow LED then you have not positioned the aircraft within the 10 degrees necessary. The completion of each positions readings is accompanied by a beep. Once the calibration is complete you will hear two beeps and the Blue LED will go out.

So what is happening? The calibration is taking several hundred readings of the three accelerometers which will be on the surface of an ellipsoid (think Australian/US football shaped) and be offset from zero. It then computes what the sensitivity scaling and offsets need to be applied to make it a sphere (think basketball) centred on zero.

Once again you only need to do this once so it is worth doing well.

Note: See Accelerometer Neutral Fine Tuning below.

** Note: Use of Analog Gyros **

There is no temperature measurement available should you choose to use external analog gyros say from your original UAVP Board. The calibration described above in Step 4 is still required but the analog gyros obtain their offset biases each flight after arming. The aircraft needs to be motionless but there is no requirement for it to be level. You may freely switch between analog gyros and the MPU6xxx gyros.

Step 5 (Magnetometer Calibration)

First it is EXTREMELY IMPORTANT that you be as far as possible away from any ferrous metal and magnetic effects so do not do this with the aircraft sitting on a bench near your computer etc.

Select the CalMag button and rotate the aircraft in all the directions you can think of including upside down and on edge. This captures the sensor offsets for the X,Y and Z axes and as with the six point accelerometer calibration fits the points to a sphere. When you start the calibration the Blue LED will start blinking and the Green LED should be on. If you see a yellow LED on then you have enough measuements with that orientation you should try another. You will get a single long beep when tha calibration is finished and the Blue LED will go off. THis can take a while if you don't get the tumpling in all directions done well enough!

You do not need to re-run the calibration unless it is clearly giving crazy answers for the computed compass heading. The offsets are stored in non-volatile memory where they are retained after disconnecting the battery.

If they are you are almost ready to fly but don't put the props on yet.

Step 6 (Motors)

This applies to multicopters only. See elsewhere fo fixed wing aircraft.

Most ESCs for multicopters do not have BEC capabilty. If yours do then make sure only one BEC is connected to the board.

We have assumed XMode so the K1 motor (ACW) is front left. Now connect the other motors:

- Rear Left K2 (CW)
- Front Right K3 (CW)
- Rear Right K4 (ACW)

If you choose to have your motors rotating in the opposite direction you can select this in the Parameters.

Note: If you plan to run camera gimbals then it is worth considering buying a high current UBEC and connecting that to an unused Rx or M connector. In that case disconnect all of the BEC centre leads and connect the UBEC to an unused M/Rx connector.

Put the aircraft on the ground. Leave the props OFF and arm the flight controller using the method chosen in the parameters (switch, roll stick, yaw stick).

You will see a dancing pattern of Yellow and Blue LEDs while the Black Box memory is cleared; this takes several seconds followed by a single beep. You should hear three starting beeps with the Red LED flashing briefly. At the end you should have Green and Red LEDs on. The Red LED means that no GPS signals are being received. UAVXGUI will speak any alarms outstanding if arming fails - button top centre of GS window.

Advance the throttle and all motors should start. Run the motors more very slowly as you can damage them if you run at high speed without props. Listening carefully move your controls and verify that if you move the aileron right that the left motor increases speed and the right motor slows down. The same for elevator. Rudder left should slow down the front/back motors and speed up the left/right motors by the same amount.

Now lift the quadrocopter up and tilt it left/right, forward/backwards. It should be obvious what the motors will do. Tipping right should cause the right motor to speed up. Again use very little throttle.

If everything seems OK you can power down and make your own decision whether it is SAFE to fit the propellors and fly.

THIS IS NOT A FLIGHT TUTORIAL SO FROM THIS POINT IT IS UP TO YOU.

Join us at:

http://www.rcgroups.com/forums/showthread.php?t=1093510

Next Steps

GPS Connection

Connect the GPS to the former Rx3 and Rx4.

If you have a GPS selected in UAVXGUI the aircraft will not fly until a stable GPS signal is received.

Accelerometer Neutral Fine Tuning

It is inevitable that the accelerometer neutrals will require small adjustments regardless of how well you leveled the aircraft. You can fine tune the neutrals using the Tx sticks. The procedure

UAVXArm32F4_V4Pinouts

Figure 2: UAVXArm32F4_V4Pinouts

is simple and is done when the aircraft is armed:

- set you Tx aileron and elevator trims to centre (no trim).
- fly the aircraft pointing away from you and note the directions that it drifts.
- land and, with the throttle closed but still armed, hold the aileron or elevator stick at maximum in direction opposite to the drift until you hear a beep - there is a delay of 2 seconds to the first beep.
- Fly again repeating until there is no drift.

You should do this when there is no wind and you only need to do it once. You should never adjust the Tx trims other than to get them as close as possible to NEUTRAL. For FW airctraft adjust the servo linkages as in any case it is bad (although now common) practice to use Tx trims.

You should never need yaw trim.

Appendices

GPS Waypoint Navigation

The most likely GPS functions you will want to use are position hold (PH) and return to home (RTH) almost always in conjunction with altitude hold (AH). Altitude hold is only active if the sticks do not move for a few seconds.

Waypoint navigation where missions are set up through the GUI Nav panel can be enabled in one of two ways.

- Aux3 to ground
- Selecting Nav with the Arm/AH/Nav channel

If there is a mission loaded (more than zero waypoints) then the centre position of the Fun/PH/RTH channel starts/resumes a mission but at the next waypoint - this is useful if you wish to abort attempts to get to the current waypoint due to wind or thermal influence.

Pin Allocations

These are the Arm processor pin assignments for the V4 Board.